

Dietary Iron Intake and Body Iron Stores Are Associated with Risk of Coronary Heart Disease in a Meta-Analysis of Prospective Cohort Studies

The link between iron intake as well as body iron stores and coronary heart disease (CHD) has been contentiously debated, and the epidemiologic evidence is inconsistent. We aimed to quantitatively summarize the literature on the association between dietary iron intake/body iron stores and CHD risk by conducting a meta-analysis of prospective cohort studies. PubMed was used to find studies published through June 2013 in peer-reviewed journals. Embase or a hand search of relevant articles was used to obtain additional articles. The pooled RRs of CHD incidence and mortality with 95% CIs were calculated by using either a random-effects or fixed-effects model, as appropriate. Twenty-one eligible studies (32 cohorts) including 292,454 participants with an average of 10.2 y of follow-up were included. Heme iron was found to be positively associated with CHD incidence (RR: 1.57; 95% CI: 1.28, 1.94), whereas total iron was inversely associated (RR: 0.85; 95% CI: 0.73, 0.999). Neither heme-iron nor total iron intakes were significantly associated with CHD mortality. Both transferrin saturation and serum iron were inversely related to CHD incidence [RR (95% CI): 0.76 (0.66, 0.88) and 0.68 (0.56, 0.82), respectively], but only transferrin saturation was inversely associated with CHD mortality (RR: 0.85; 95% CI: 0.73, 0.99). In conclusion, total iron intake and serum iron concentrations were inversely associated with CHD incidence, but heme iron intake was positively related to CHD incidence. Elevated serum transferrin saturation concentration was inversely associated with both CHD incidence and mortality. Future research is needed to establish the causal relation and to elucidate potential mechanisms.